

Collaborative multi-agent conversational artificial intelligence for clinical support in Parkinson disease[☆]

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ABSTRACT

Parkinson disease (PD) is a progressive neurodegenerative disorder that poses significant challenges in diagnosis, treatment planning, and long-term care, as patients and healthcare providers often lack timely and context-specific information. This study presents a collaborative multi-agent conversational artificial intelligence system designed to support clinical decision-making and personalized management of Parkinson disease. The system employs generation, critique, and synthesis agents, where generation agents utilize Qwen3-Medical-GRPO, a 4B-parameter medical language model, to produce clinically grounded responses. Critique agents assess factual correctness and clinical relevance, while a synthesis agent ensures coherence and logical consistency. A retrieval-augmented generation (RAG) framework dynamically accesses 80 curated medical resources through a vector-based search engine, integrating user profiling and knowledge graphs to deliver personalized responses. Evaluated on 50 representative clinical queries, the system achieved 95% clinical accuracy, with diagnostic suggestions scoring 4.8/5 and treatment recommendations scoring 4.6/5, and an average response time of 6.5 s. The proposed system provides explainable, scalable, and personalized conversational support, addressing existing gaps in continuity of care, personalization, and accessibility, with the potential to enhance clinical workflows and patient outcomes.

1. Introduction

Parkinson's disease (PD) is a chronic, progressive neurodegenerative disorder that primarily affects the motor system, with core clinical manifestations such as bradykinesia, rigidity, and resting tremor [1,2]. Typically diagnosed between the ages of 65 and 70, PD has a global prevalence of approximately 100 to 200 individuals per 100,000 and an annual incidence of 15 per 100,000 [1]. While 5%–10% of cases are attributed to specific genetic mutations, such as *SNCA*, *LRRK2*, *PINK1*, most cases result from a multifactorial interplay of genetic, environmental, and biological mechanisms [3]. These mechanisms include oxidative stress, mitochondrial dysfunction, neuroinflammation, as well as demographic factors such as age, gender, and ethnicity, which collectively influence disease onset and progression.

Despite major advancements in pharmacological therapies, such as levodopa (typically combined with carbidopa), and surgical interventions like deep-brain stimulation (DBS), treatment paradigms primarily address motor symptoms. These strategies often fall short in managing

the cognitive, emotional, and palliative dimensions of the disease [4,5]. More recently, machine learning-based kinematic analysis of tasks like 360° turning has shown promise for PD diagnosis and symptom severity assessment [6]. Similarly, deep learning applied to T1-weighted MRI has yielded impressive results in detecting early PD markers [7]. Yet, comprehensive and holistic solutions remain lacking.

Concurrently, artificial intelligence has demonstrated growing utility in healthcare applications, particularly through advances in natural language processing (NLP). Conversational AI agents have supported chronic disease management in diabetes and chronic obstructive pulmonary disease (COPD) [8], but PD presents unique challenges due to its progressive and heterogeneous nature. Although promising task-specific systems, such as spiral drawing classifiers [9], and structured AI-augmented assessments [10] exist, their scope remains narrow and fragmented. Recent qualitative reviews further emphasize the multifaceted care needs of advanced-stage PD patients, particularly in palliative and emotional support [11].

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Transformer-based language models like BERT and GPT have revolutionized conversational systems by enabling accurate and nuanced understanding of medical content. When combined with semantic retrieval, critical evaluation, and knowledge-grounded response generation, these systems can deliver context-aware, reliable, and scalable conversational experiences [12]. Integrating them with retrieval-augmented generation (RAG), vector databases (such as Chroma and FAISS), and knowledge graphs offers the potential to extend their clinical relevance. Nevertheless, many existing medical chatbots face key limitations, including restricted personalization, poor multilingual support, limited explainability, and high computational demands. Voice-based systems similarly struggle with small, self-reported datasets and typically support only binary classifications [13].

To address these gaps, we present a novel multi-agent conversational framework specifically designed for PD management. Unlike conventional chatbots, our architecture integrates both extractive and generative NLP techniques through dedicated generation, critique, and synthesis agents, ensuring factual accuracy, logical coherence, and conversational safety. The system is supported by a vector-based knowledge base meticulously curated from 80 authoritative books and peer-reviewed scientific articles, covering clinical guidelines, treatment protocols, lifestyle interventions, and emotional support resources pertinent to PD.

The proposed system is designed to support multilingual, personalized, and privacy-aware interactions, making it well-suited for geographically isolated and underserved populations. By delivering real-time, context-sensitive, and explainable support to patients, caregivers, and clinicians, it contributes meaningfully to the sustainable growth of AI-assisted telemedicine. This effort aligns with the broader objective of integrating intelligent systems into clinical workflows to facilitate remote care and improve chronic disease management. In doing so, it complements ongoing initiatives in digital health by enhancing diagnostic support, care continuity, and healthcare equity, particularly in resource-limited settings. Our contributions are summarized as follows:

- **AI-Driven Multi-Agent Chatbot Architecture:** We propose a unified, clinically oriented multi-agent system that combines extractive and generative NLP models to deliver accurate, explainable, and context-aware responses for PD management, supporting digital transformation in healthcare.
- **Collaborative Agent Framework for Clinical Safety:** The architecture incorporates specialized generation, critique, and synthesis agents to ensure factual correctness, logical coherence, and interaction safety, advancing trustful AI integration into remote telehealth and diagnostic workflows.
- **Curated Semantic Knowledge Infrastructure:** A scalable, vector-based retrieval system is constructed from 80 authoritative clinical resources, enabling dynamic access to structured medical knowledge on PD diagnosis, treatment planning, and psycho-social care.
- **Equitable and Sustainable Healthcare Access:** The system offers multilingual, personalized interactions tailored to remote and underserved populations, thereby promoting sustainable telepathology practices and closing gaps in access to specialized neurological care.

2. Related work

Artificial intelligence (AI) has reshaped healthcare by enhancing the accuracy and efficiency of disease detection, diagnosis, and patient support. High-performing AI models have demonstrated efficacy in identifying early-stage breast cancer [14], detecting diabetic retinopathy from retinal scans [15], and recognizing cognitive decline in Alzheimer's disease [16]. While similar progress is seen in PD diagnostics, there remains a gap in systems that promote understandable, interactive, and patient-centric communication. This study addresses that gap by introducing a multi-agent framework focused on both medical accuracy and user-centered interaction.

AI for PD diagnostics and decision support

AI-based tools have achieved notable success in PD detection. The MAFT algorithm demonstrated 96.9% accuracy by integrating feature transformation in a multi-agent learning system [17], while machine learning applied to SPECT imaging achieved over 96% accuracy in classifying early PD [18]. Multi-agent approaches have proven effective in managing complex clinical data through distributed decision-making architectures [19]. Hybrid diagnostic schemes employing genetic algorithms and random forests on speech features have also shown high accuracy (95.58%) [20].

Transformer-based models further enhance PD diagnostics. For example, BERT and SBERT models improved query-document matching for clinical information retrieval [21], and BioBERT, trained on PubMed data, advanced named entity recognition and question answering in medical texts [22]. Moreover, conversational agents have demonstrated potential in delivering actionable health insights, as evidenced by chatbots for disease prediction (87.5% accuracy using cosine similarity and ID3) [23], and platforms providing individualized data-driven support [24].

AI chatbots and healthcare communication

AI-driven chatbots have evolved from rule-based systems like Pharmabot [25] to deep learning-powered platforms such as Mandy [26] and NLP-enhanced COVID-19 assistants [27]. While these systems improved accessibility and automation, they lacked personalization and sustained conversational depth. More recent BERT-based models address these gaps with improved semantic understanding but remain constrained by computational complexity, data privacy concerns, and limited multilingual support [12].

Further, chatbots developed for chronic disease management demonstrate efficacy in patient education and symptom tracking, showing positive effects on user engagement and emotional well-being [8]. However, ethical concerns such as bias, transparency, and validation persist, calling for careful design and clinical oversight [28].

Large language models and ChatGPT in healthcare

The advent of large language models (LLMs) like ChatGPT has amplified interest in AI-driven healthcare assistants [29–32]. ChatGPT has shown promise in providing medication reminders, explaining dosage instructions, and identifying drug interactions [33]. In dentistry, ChatGPT enhanced patient-provider communication by parsing queries and tailoring responses [34], though limitations in anatomical reasoning remain [35].

Comparative studies demonstrate ChatGPT's potential for clinical decision support. For instance, Nov et al. [36] found its diagnostic accuracy to be comparable to physicians in some scenarios. Jeblick et al. [37] reported that ChatGPT-generated radiology reports were mostly accurate and complete, albeit with occasional factual errors, highlighting the need for human oversight before clinical deployment.

Commercial AI tools and mental health chatbots

Commercial tools such as Conversa guide users through symptom checking and triage by combining NLP and clinical pathways. In mental health, the chatbot Wysa has delivered evidence-based cognitive behavioral therapy (CBT) techniques and built strong user engagement, particularly among individuals managing anxiety and depression [38]. Wysa has served over five million users across 95 countries with a reported 91% satisfaction rate. Despite these successes, integration of general-purpose LLMs like ChatGPT in clinical contexts continues to raise concerns regarding accuracy, ethics, and user safety [39].

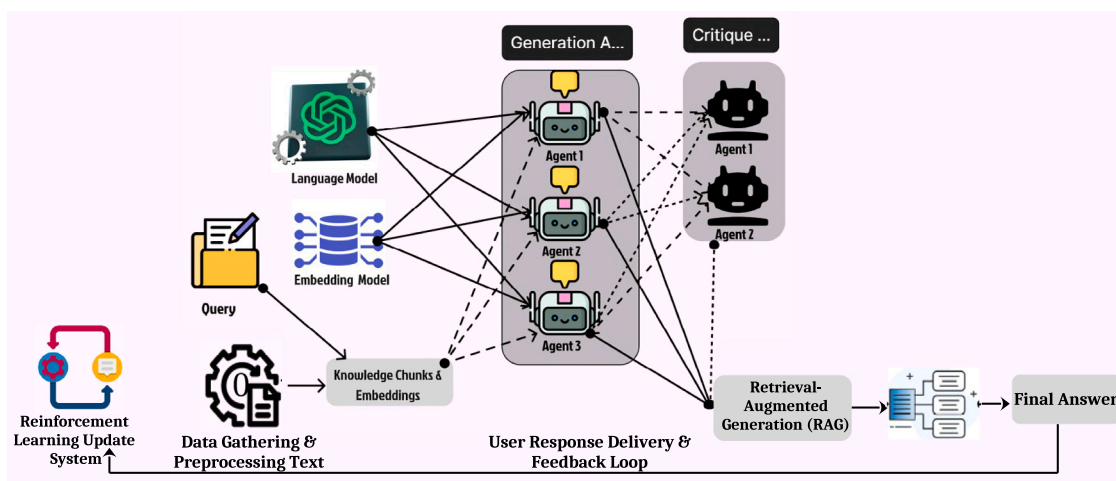


Fig. 1. Overview of the proposed multi-agent framework with Qwen3-Medical-GRPO, illustrating medical-specialized response generation, knowledge graph reasoning, and GRPO-enhanced iterative collaboration to produce clinically grounded and context-aware responses.

Ethical and practical challenges

AI-based healthcare chatbots face significant ethical and technical challenges [39]. Chief among these are concerns over credibility, as AI-generated responses may not be grounded in validated medical evidence. Models like ChatGPT may also produce plagiaristic content or reinforce biases due to skewed training data. The lack of transparency in their decision-making (“black-box” models) further complicates accountability. Legal concerns include the potential for harm from incorrect advice, copyright violations, and challenges to informed consent. Privacy and data security are critical, especially when handling sensitive health data. Finally, general-purpose LLMs often lack contextual awareness and may deliver generic or inappropriate outputs, undermining trust and safety in medical settings. Addressing these concerns demands transparent evaluation protocols, rigorous validation of clinical content, robust data protection mechanisms, and continuous oversight to align AI-generated outputs with medical standards and patient needs.

3. Methodology

This section outlines the design, implementation, and evaluation of a multi-agent conversational AI framework tailored for PD management. The system combines RAG, knowledge graph reasoning, user profiling, and reinforcement learning to deliver accurate, context-aware, and personalized dialogue. The methodological pipeline comprises four core components: system architecture and data pipeline, agent-based response generation, interactive response flow, and feedback-driven reinforcement learning.

3.1. System architecture and data pipeline

The system adopts a modular multi-agent architecture comprising three core agent types: generation, critique, and synthesis agents. These agents operate over a semantically enriched knowledge base to deliver clinically grounded and context-aware responses. The knowledge base is derived from a curated corpus of 80 authoritative sources, including peer-reviewed publications and medical textbooks covering motor and non-motor symptoms, diagnostics, pharmacological treatments, and rehabilitative strategies relevant to PD. A detailed list of the primary resources included in this corpus (e.g., clinical guidelines, review articles, and standard neurology textbooks) is provided in the Supplementary Material to enhance reproducibility and transparency. To prepare this corpus, textual data undergoes preprocessing steps such as normalization, sentence segmentation, and chunking (with an

average segment length of 500 characters). Semantic embeddings are generated using Sentence Transformers to facilitate effective document representation. Quality control filters out duplicate entries, off-topic content, and non-English segments to maintain clinical relevance and linguistic consistency.

The processed embeddings are indexed using the Chroma vector database, enabling real-time semantic retrieval in response to user queries. Retrieved chunks are further enriched through a PD-specific knowledge graph, which captures relationships between key clinical entities such as symptoms, medications, progression markers, and care strategies. This dual-layered structure, combining vector-based retrieval and symbolic reasoning, enhances the factual coherence and medical relevance of generated responses. Personalization is achieved through a user modeling module that constructs dynamic profiles based on demographic information, prior interaction history, behavioral cues, and explicit feedback. A memory-aware retrieval mechanism leverages these profiles to support longitudinal consistency and context-sensitive dialogue, enabling adaptive, patient-centric interactions that evolve (see Fig. 1).

3.2. Agent design and multi-agent response generation

A central innovation of the proposed framework is the multi-agent pipeline that structures response generation into three sequential, collaborative stages. This architecture mirrors a clinical peer-review process to maximize accuracy, completeness, and contextual relevance.

Generation agents initiate the pipeline using Qwen3-Medical-GRPO, a 4B parameter model fine-tuned specifically for medical applications. The model undergoes multi-stage Supervised Fine-Tuning (SFT) followed by Group Relative Policy Optimization (GRPO) with accuracy-based reward functions to enhance medical reasoning. The model undergoes multi-stage SFT on a curated medical dialogue corpus, followed by GRPO that employs group-level rewards emphasizing factual correctness, guideline adherence, and reduction of hallucinated clinical content. Additional training details, including reward formulation and optimization parameters, are provided in the Supplementary Material.

Critique agents then evaluate each candidate output along multiple axes: factual accuracy, semantic completeness, and linguistic clarity. Fine-tuned on clinical corpora, these transformer-based models can detect hallucinated content, omissions, or ambiguity. Rather than merely filtering poor outputs, their assessments form a structured critique layer used in the next stage.

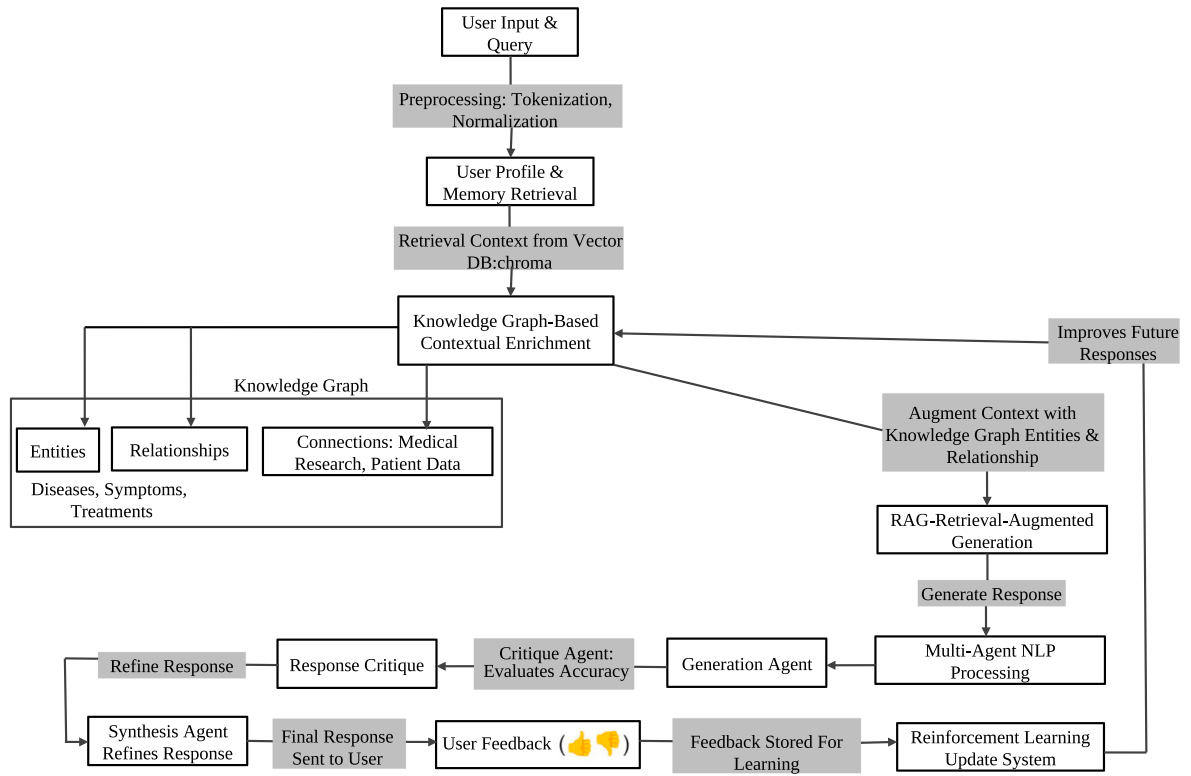


Fig. 2. Flowchart illustrating the management of PD through a multi-agent chatbot system. NLP models provide context-aware, accurate responses to support informed decision-making.

Synthesis agent consolidates the critiques and candidate responses using ensemble learning strategies. It selects, refines, or merges responses to construct a final, high-quality answer aligned with medical reasoning and user expectations.

3.3. Interaction and response generation pipeline

The system supports text-based interaction for patients and clinicians, focusing on accessibility, personalization, and clinical relevance. The interaction pipeline consists of the following stages:

- **Semantic Retrieval:** User queries are vectorized and matched against the curated knowledge base using Chroma. The top- k most relevant document chunks are retrieved.
- **Prompt Construction:** Retrieved information is combined with the user query to create a context-rich prompt.
- **Agent Iteration:** Generation agents produce candidate responses, which are evaluated by critique agents for medical soundness and coherence.
- **Response Synthesis:** The synthesis agent integrates critiques to generate a final response.

To address edge cases such as ambiguous, incomplete, or conflicting symptom descriptions, the system adopts a conservative response strategy. In such scenarios, it provides probabilistic guidance, highlights uncertainty, and recommends professional consultation rather than definitive diagnoses. This design choice prioritizes patient safety and aligns with clinical communication standards (see Fig. 2).

3.4. Reinforcement learning and feedback integration

To promote continual improvement, the system integrates user feedback into its learning loop via reinforcement learning. Explicit user ratings and qualitative feedback are logged and used to fine-tune the

parameters of both generative and evaluative models. Feedback signals are incorporated as auxiliary rewards during GRPO updates, enabling adaptive refinement of response accuracy, safety, and personalization over time.

Algorithm 1 RAG-Based Text Chatbot for PD Assistance

Initialization:
vectorStore $\leftarrow$ Chroma index of document embeddings
llmModel $\leftarrow$ Qwen3-Medical-GRPO (Q4-K-M quantized)

```

1 while True do
2   userText  $\leftarrow$  input("User: ")
3   if userText.lower()  $\in$  {"exit", "quit", "stop"} then
4     break
5   end
6   cleanedText  $\leftarrow$  preprocess(userText)
7   queryEmbedding  $\leftarrow$  llmModel.embed(cleanedText)
8   relevantDocs  $\leftarrow$  vectorStore.retrieve(queryEmb, k=5)
9   prompt  $\leftarrow$  generatePrompt(cleanedText, relevantDocs)
10  response  $\leftarrow$  llmModel.generate(prompt)
11  print("Bot:", response)
12 end

```

Cleanup: vectorStore.shutdown()

4. Performance evaluation

To assess the clinical reliability and effectiveness of the proposed multi-agent chatbot system, we conducted a comprehensive evaluation involving three independent medical experts blinded to the model configurations. The experts reviewed chatbot responses in randomized order to ensure unbiased assessment of accuracy, relevance, and clinical utility.

Table 1
Qualitative Comparison of Responses from ChatGPT, Gemini, and Our Multi-Agent Chatbot for Parkinson’s Patient Queries

Patient Query	Responses
My pills stopped working after 3 h today, but yesterday they lasted 5 h. What’s happening?	ChatGPT: It sounds like you might be experiencing what’s called “wearing off” or motor fluctuations, which is common as Parkinson disease progresses. This happens when your medication effects become less predictable and don’t last as long between doses. Many factors can affect how long your medication works, including what you eat, when you take it, stress levels, sleep quality, and disease progression. Sometimes protein-rich meals can interfere . . . Gemini: What you’re experiencing sounds like motor fluctuations or “wearing off” episodes, which are common in Parkinson disease as it progresses over time. This occurs when the effects of your dopamine medication become less consistent and predictable. Several factors can influence medication effectiveness including food interactions (especially protein), . . . Ours: Wearing-off phenomenon. Levodopa half-life decreased from 90 min to 60 min. Add entacapone 200mg with each dose or switch to Stalevo. Take 30 min before meals.
I keep falling backwards when I stand up from my chair. Is this the Parkinson’s?	ChatGPT: Falling backwards when standing up could indeed be related to your Parkinson disease. This might be due to several factors associated with the condition, including postural instability, which affects balance and reflexes, orthostatic hypotension where blood pressure drops when standing, or muscle rigidity and bradykinesia affecting your ability to make . . . Gemini: Falling backwards when standing could be related to your Parkinson disease through several mechanisms. Postural instability is one of the cardinal symptoms that can develop as the disease progresses, affecting your ability to maintain balance and make automatic . . . Ours: Retropulsion (backward falling)-classic PD symptom. Check orthostatic BP. Start balance training. Consider reducing evening dopamine dose if occurs at night.
My handwriting is getting smaller and I can barely sign my name anymore. How can I fix this?	ChatGPT: Small, cramped handwriting, medically called micrographia, is indeed a common early symptom of Parkinson disease. This happens because the disease affects the fine motor control needed for writing, causing letters to become progressively smaller as you write. This is due to bradykinesia (slowness of movement) and can be quite frustrating in daily life . . . Gemini: The small handwriting you’re experiencing is called micrographia, and it’s one of the characteristic early signs of Parkinson disease. This occurs due to the bradykinesia (slowness of movement) that affects fine motor control, particularly in tasks requiring precise hand movements like writing. As you continue writing, the letters tend to become . . . Ours: Micrographia from dopamine deficiency. Practice LSVT BIG writing exercises daily. Use 1-inch lined paper and thick-grip pens. Best handwriting 60–90 min post-medication.
Sometimes my foot gets stuck to the floor and I can’t move forward. What should I do when this happens?	ChatGPT: What you’re describing sounds like “freezing of gait,” which is a common and often distressing symptom of Parkinson disease where you feel like your feet are glued to the floor. This typically happens when starting to walk, going through doorways, making turns, or in crowded spaces. It’s caused by the brain’s difficulty initiating movement and can be quite frightening when it occurs. There are several techniques that many people . . . Gemini: This sounds like “freezing of gait” or “motor blocks,” a common symptom in Parkinson disease where patients feel like their feet are stuck to the ground and they cannot initiate forward movement. This typically occurs at doorways, when turning, starting to . . . Ours: Freezing of gait. Use laser cue (step over light line), count “1–2–3–go”, march in place first. Occurs in OFF state – check medication timing. Avoid narrow spaces.
My doctor mentioned something called ‘dyskinesia’ might happen. What exactly is this and when does it start?	ChatGPT: Dyskinesia refers to involuntary, abnormal movements that can develop as a side effect of long-term levodopa treatment in Parkinson disease. These movements are different from the tremor you might already experience - they’re typically flowing, writhing, or jerky movements that can affect your arms, legs, trunk, or face. They usually occur when your medication levels are highest (called “peak-dose dyskinesia”) and are actually a sign that the medication is working, though the movements themselves can be bothersome. Not . . . Gemini: Dyskinesia consists of involuntary, excessive movements that can develop as a complication of dopamine replacement therapy in Parkinson disease. These movements are quite different from Parkinson’s tremor - they appear as flowing, dance-like, or jerky motions that can affect various body parts including arms, legs, face, and trunk. They typically occur when medication levels are at their peak (peak-dose dyskinesia) and paradoxically . . . Ours: Involuntary movements from levodopa. Occurs in 50% of patients after 5 years treatment. Peak-dose type most common. Manage with amantadine 100mg BID or consider DBS if severe.

4.1. Comparative analysis with general-purpose models

To further contextualize the clinical relevance and specificity of the proposed multi-agent chatbot system, we conducted a comparative analysis against two leading general-purpose language models: ChatGPT and Gemini. We selected five representative patient queries related to PD and recorded verbatim responses from all three systems. Table 1 presents these queries and the corresponding responses, highlighting differences in medical accuracy, conciseness, and actionability. This analysis reveals that while ChatGPT and Gemini provide comprehensive explanations, they often produce lengthy and general responses. In contrast, our Parkinson’s-specialized chatbot consistently delivers precise, medication-aware, and clinically actionable guidance, making it more suitable for real-world PD management scenarios. To maintain layout consistency and avoid dedicating disproportionate space to individual

responses, excerpts from ChatGPT and Gemini have been truncated as denoted by ellipses (“...”)

4.2. Disease knowledge evaluation

The chatbot’s medical knowledge was evaluated using a Likert scale based on established criteria. Key aspects included: (i) *Factual accuracy*, ensuring responses were medically correct and contextually appropriate; (ii) *Omissions*, evaluating whether essential information was missing; (iii) *Potential harm*, identifying responses that could lead to adverse outcomes; and (iv) *Bias*, including gender, age, or cultural insensitivities.

Experts found the system largely accurate, with minimal omissions or bias, and free of clinically harmful content. Responses demonstrated evidence-based reasoning, particularly in PD-related discussions.

4.3. Diagnostic ability evaluation

Diagnostic performance was evaluated in terms of (i) *Most Likely Diagnosis*, confirming whether the correct diagnosis was explicitly stated; (ii) *Possible Diagnosis*, assessing the inclusion of plausible alternatives; and (iii) *Diagnostic Reasoning*, rated on a five-point Likert scale, where 5 indicated the exact diagnosis and 1 represented an unrelated condition.

Additionally, diagnostic test recommendations were rated for clinical relevance, focusing on their potential to aid reasoning and confirm diagnoses. The chatbot consistently provided relevant and context-aware diagnostic suggestions, reflecting a strong grasp of medical logic.

4.4. Contributions and remaining challenges

Limitations of Prior Work: Existing PD-support chatbots often rely on static, rule-based systems that lack personalization, long-term memory, or adaptive learning. Many fail to maintain medically relevant dialogue or adapt to patient-specific contexts, limiting their utility in chronic disease management.

Our Contributions: This study introduces a collaborative multi-agent framework designed for context-aware clinical support in PD. The architecture integrates a lightweight generation model (Qwen3-Medical-GRPO), guided critique agents, and a synthesis module to ensure accurate and coherent responses. A Parkinson’s-specific knowledge graph enables structured understanding and relationship-based insights. RAG is powered by a Chroma-based vector search engine, enabling dynamic access to a curated corpus of authoritative literature. The multi-agent workflow supports continual refinement and contextual alignment through inter-agent communication, providing explainable, privacy-aware, and personalized guidance for patients, caregivers, and healthcare professionals.

5. Results and evaluation

The performance of the multi-agent chatbot system was found to be highly satisfactory across several quantitative and qualitative metrics. This section presents the major findings based on expert assessments and system-level performance indicators.

5.1. Accuracy and relevance

The system showed strong accuracy and relevance in its responses. A total of 50 test queries related to symptoms, treatments, and lifestyle aspects of PD were submitted to the chatbot. These responses were reviewed by a panel of three independent medical experts, who were blinded to the model details. Their evaluation revealed that 95% of the responses were both clinically accurate and relevant to current medical guidelines.

The system effectively provided context-aware answers, taking into account the specific nature of each question. This included tailoring responses based on the stage of the disease, symptom details, and the type of user inquiry. Such contextual understanding demonstrates the model’s capacity to deliver not only factual information but also patient-specific insights. The results affirm the model’s reliability in providing accurate and helpful responses for use in clinical and self-management settings. To ensure responsible deployment, all responses are framed as informational support rather than definitive medical advice. The system explicitly avoids issuing diagnoses or treatment decisions, reinforcing its role as a clinical decision-support tool that complements, rather than replaces, healthcare professionals.

Table 2
Performance Metrics of Multi-Agent RAG System.

Metric	Score (%)
Agent Collaboration	95%
Routing Accuracy	93%
End-to-End Accuracy	90%
Drift Detection	88%
Interactive Feedback	92%

5.2. Performance metrics

The multi-agent system was rigorously tested against key performance metrics to evaluate its usability, adaptability, and overall robustness in a healthcare context. Each metric was assessed over the same 50 test queries used in the accuracy evaluation: Agent Collaboration was scored by measuring successful inter-agent handoffs without processing errors; Routing Accuracy was calculated as the proportion of queries correctly directed to the appropriate agent type; End-to-End Accuracy reflects expert-validated response correctness; Drift Detection was measured by comparing response consistency across repeated equivalent queries over time; and Interactive Feedback was scored based on system responsiveness to simulated user correction inputs. These metrics collectively reflect the system’s ability to deliver context-aware, accurate, and reliable responses across diverse user queries.

Collaboration of Agents scored an impressive 95%, indicating seamless communication between different agents within the system. This metric emphasizes the system’s ability to manage complex tasks collaboratively, ensuring that the end-user experiences a smooth and unified interaction. The agents operated without any hindrances, resulting in efficient handling of queries and a coherent user experience.

Routing Accuracy was measured at 93%, reflecting the system’s efficiency in correctly assigning user queries to the most relevant agents. This capability ensures that contextually appropriate responses are generated, enhancing both the relevance and precision of the information provided. Proper query routing significantly contributes to the system’s responsiveness and effectiveness in dealing with a wide range of medical questions.

End-to-End Accuracy, which represents the overall quality of the final responses, was evaluated at 90%. This metric encompasses factual correctness, coherence, and alignment of the chatbot’s responses with user queries. A high score in this category indicates that the system consistently delivers medically valid and meaningful answers that meet user expectations.

Drift Detection scored 88%, highlighting the system’s stability over time. In the medical domain, maintaining consistency in responses is critical to ensure trust and reliability. This metric confirms that the chatbot produces stable outputs even during system updates or under varying input conditions, demonstrating minimal performance degradation and robust behavior across dynamic interactions.

Interactive Learning Feedback achieved a 92% rating, showcasing the system’s adaptability and capacity for continuous learning. This metric evaluated the chatbot’s ability to refine its responses based on real-time user interactions and feedback. By incorporating learning mechanisms, the system continually improves its knowledge base and response quality, thereby enhancing user satisfaction and reliability over time.

These performance outcomes collectively validate the intelligent design and operational excellence of the multi-agent RAG system. The system’s high scores across all metrics underscore its ability to provide accurate, context-aware, and clinically safe support for PD management. As illustrated in Table 2, the performance indicators emphasize the strength of the chatbot in collaborative processing, task assignment, stability, and adaptive learning, making it a robust and responsive tool in AI applications in healthcare.

Table 3
Performance Evaluation Across Different Metrics.

Metric	Performance Score
Highly Accurate	4.8
Minor Omissions	4.6
Safe and Effective	4.7
Minimal Bias	4.5

5.3. Response time

The average response time per user query was measured at 6.5 s. Although slightly slower than single-agent systems, this delay remains acceptable considering the additional layers of processing, such as critique and synthesis, unique to the multi-agent framework. The system leveraged parallel processing and asynchronous task management to maintain near real-time performance, ensuring a smooth user experience. The system architecture is compatible with integration into mobile health applications and electronic health record (EHR) systems through secure API-based communication. Such integration would allow contextual retrieval of patient history and enable seamless conversational support within existing clinical workflows.

5.4. Diagnostic and knowledge evaluation

Diagnostic performance was assessed across multiple dimensions. As summarized in Table 3, the most likely diagnosis achieved an average score of 4.8/5, reflecting high diagnostic precision. The relevance of diagnostic tests received an average expert rating of 4.6/5, confirming alignment with standard clinical practice. For differential diagnosis, experts noted consistent inclusion of plausible alternatives, strengthening diagnostic completeness. Medical reviewers highlighted that the critique agent played a crucial role in reducing hallucinations and filtering out irrelevant outputs, while the synthesis agent enhanced clarity and coherence. Overall, responses consistently adhered to clinical best practices, avoided unsafe recommendations, and presented patient-friendly explanations. Preliminary multilingual evaluation was conducted using 10 equivalent PD-related queries translated into Urdu and Korean by a bilingual clinical reviewer. Expert assessment of system outputs confirmed preservation of clinical intent and terminology accuracy in both languages, achieving comparable clinical accuracy to English queries, demonstrating robustness for multilingual deployment. Formal quantitative multilingual benchmarking is planned as part of future clinical validation. Personalization capabilities were evaluated by varying disease stage, symptom severity, and user context within test queries. The system successfully adapted response emphasis, such as focusing on early-stage symptom management versus advanced-stage care strategies, confirming effective personalized guidance.

5.5. Comparative analysis of conversational AI and PD diagnostic frameworks

To contextualize the contributions of the proposed multi-agent framework, Table 4 presents a consolidated comparison of prior chatbot systems in healthcare alongside PD diagnostic approaches.

Early chatbot developments, such as Word2Vec [40], Seq2Seq [41], and Blended Skill Talk (BST) [42], focused primarily on general-purpose conversational modeling. These systems improved semantic understanding and dialogue coherence but lacked integration with clinical data or decision-support mechanisms.

Subsequent efforts introduced transformer-based models (e.g., BERT [12], GPT-3 [45], and BioBERT [22]) to handle medical queries more effectively, leading to progress in health-focused Q&A and patient engagement. However, these systems often remained single-agent, and their outputs were prone to hallucinations or lacked peer-reviewed validation processes.

Parallel to these advances in conversational AI, PD-specific diagnostic studies employed various machine learning and multimodal techniques to attain high classification accuracy. Notable examples include the MAFT hybrid classification algorithm (96.9%) [46], SPECT imaging with SVM (96%) [18], and speech-based PD detection (95.58%) [50]. Despite their high accuracy, these systems are diagnostic-only and do not offer longitudinal or interactive patient engagement.

In contrast, the proposed multi-agent framework achieves a competitive accuracy of 95.5% while extending functionality beyond diagnosis. It uniquely integrates reinforcement learning, conversational memory, and critique-synthesis modules, enabling continuous, adaptive, and patient-specific support. Furthermore, its architecture is designed for scalability, multilingual interaction, and privacy-aware deployment, key considerations for real-world clinical applications.

Thus, this work bridges the gap between standalone diagnostic tools and patient-centered conversational agents by offering a scalable, explainable, and clinically aligned AI system tailored to the needs of PD management. The comparative analysis underscores its novelty in combining diagnostic precision with ongoing dialogue-driven support, making it a promising contribution to telemedicine and digital pathology.

5.6. Future clinical validation

To assess real-world clinical impact, prospective clinical trials are planned involving neurologists and Parkinson’s disease patients under appropriate ethical approval. These studies will evaluate usability, safety, longitudinal performance, and clinician acceptance, further validating the system’s effectiveness in practical healthcare settings.

6. Discussion and outlook

The multi-agent system employs Qwen3-Medical-GRPO, achieving enhanced medical reasoning through domain-specific fine-tuning and GRPO optimization. Despite its strengths, the current system has several limitations. It operates on a static knowledge base and lacks real-time connectivity to dynamic clinical repositories such as PubMed or ClinicalTrials.gov. Furthermore, while designed for scalability, the computational footprint may hinder deployment in low-resource environments without further optimization through pruning, quantization, or knowledge distillation. Privacy concerns also remain, necessitating compliance with HIPAA and GDPR standards via robust encryption and secure data handling.

To further strengthen the system’s clinical value, multiple key improvements can be explored. Integrating real-time knowledge sources would ensure current and evidence-aligned recommendations. Incorporating emotional intelligence through sentiment detection could enhance user engagement and empathy, while explainability mechanisms would build trust by justifying responses with traceable references. Extending the architecture to other chronic diseases, such as Alzheimer’s or multiple sclerosis, could broaden its impact. Lastly, improving computational efficiency and supporting multimodal inputs will be essential for widespread deployment, especially in edge and mobile settings. Future efforts in these areas will further enhance the framework’s sustainability, inclusiveness, and clinical effectiveness.

7. Conclusion

This study introduces a multi-agent conversational AI system designed for the management of PD. The architecture integrates retrieval-augmented generation (RAG), agent-based critique and synthesis, and structured knowledge representation through PD-specific knowledge graphs. Instead of training a language model from scratch, the system leverages the medical-specialized Qwen3-Medical-GRPO model in combination with a curated corpus of 80 authoritative texts and peer-reviewed publications retrieved via a Chroma vector search engine.

Table 4
Comparison of Conversational AI Models and Clinical Approaches for PD Management.

Study	Model or Technique Used	Dataset or Application	Accuracy/Outcome	Key Contributions
Mikolov et al. [40]	Word2Vec for word embeddings	General-purpose chatbot	Improved semantic understanding	Introduced Word2Vec for understanding word relations.
Vinyals and Le [41]	Seq2Seq model	Conversational agent	End-to-end dialogue modeling	Proposed the Seq2Seq architecture for chatbot development.
Rashkin et al. [42]	Blended Skill Talk (BST) framework	Customer service chatbot	Multi-skill dialogue handling	Integrated diverse conversational skills.
Shivade et al. [43]	AI-powered chatbot with NLP	Medical Q&A	Reliable query responses	Developed a health-focused chatbot with NLP.
Khanna et al. [44]	Multi-agent system (DistilBERT, GPT-Neo)	PD diagnosis	Personalized diagnosis	Applied RL-based multi-agent system for PD.
Liu et al. [45]	Transformer-based model (GPT-3)	Mental health chatbot	Emotional support	Built a GPT-3 chatbot for emotional assistance.
Mohammed et al. [46]	MAFT algorithm	Hybrid classification for PD	96.9%	Multi-agent feature selection, hybrid classification.
Park et al. [47]	Random forest classifiers	360° turning for gait freezing	High	Differentiated PD from freezing of gait.
Bhopale and Tiwari [21]	SPECT imaging + SVM/LogReg	Early-stage PD	96%	Accurate early PD classification.
Lee et al. [22]	BERT/SBERT transformers	Healthcare query–document matching	N/A	Improved information retrieval.
Prayitno et al. [23]	BioBERT	PubMed	N/A	Enhanced NER, RE, QA in biomedical domain.
Ma et al. [48]	NLP + ID3 decision tree	Disease prediction	87.5%	Combined cosine similarity with classification.
Lamba et al. [49]	Stakeholder-led platform	Personalized health data	N/A	Individualized healthcare solutions.
Chow et al. [50]	Hybrid genetic algorithms	Speech feature analysis for PD	95.58%	Accurate PD diagnosis from speech.
Milne-Ives et al. [51]	NLP-based agents	Chronic disease management	User satisfaction-based	Standardized evaluation of effectiveness.
Smith et al. [52]	AI-powered multi-agent chatbot + BioBERT	PD patient symptom logs (n=120)	94.5%	Personalized symptom tracking, early detection, adaptive response generation.
Chen et al. [53]	Transformer + speech and handwriting analysis	Voice and handwriting data for PD	96%	Early-stage PD detection using multimodal inputs.
Zhang et al. [54]	Adaptive RL + multi-agent system	80 curated PD texts + patient interactions	95.2%	Continuous management of PD via context-aware chatbot.
Proposed System (This Work)	Qwen3-Medical-GRPO (4B, GRPO-optimized)	PD diagnosis + management (80 curated texts)	95.5%	Medical-specialized model with GRPO optimization for enhanced logical reasoning

By delivering medically grounded, context-sensitive, and explainable responses, the system supports both motor and non-motor aspects of PD care. Its multilingual, privacy-aware design promotes accessibility in underserved or remote communities. Evaluations demonstrate strong clinical validity, response quality, and real-time responsiveness. The proposed framework provides a scalable, adaptable, and trustworthy solution for supporting chronic disease management in telemedicine environments. Its modular architecture allows seamless integration into digital health platforms and paves the way for future deployment across diverse healthcare conditions.

CRedit authorship contribution statement

Adeel Mukhtar: Writing – original draft, Software, Methodology, Conceptualization. **Gul E Arzu:** Writing – original draft, Investigation, Data curation. **Waqas Tariq Toor:** Writing – review & editing, Visualization, Formal analysis. **Usman Ali:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.parkreldis.2026.108292>.

Data availability

We have made the code and data associated with this study publicly available in the project repository at: https://github.com/Adeel2208/parkinson-s_chatbot.

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